Tackling Competition Issues with Advanced Methods

E.CA Interview

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BiGSEM Bielefeld Graduate School of Economics and Management

EPOC Economic Policy in Complex Environments

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My Research: Five Papers

Research Portfolio

- **P1** -- **Data Restrictions & Competition:** How data privacy regulations shape firm strategies and market structures in software markets
- **P2** -- **Capacity & Network Effects:** Strategic service capacity investment under congestion and consumer networks in ICT markets (Job Market Paper)
- **P3** -- **ABM Calibration:** Parameter estimation methods using real telecom data (AT&T, T-Mobile)
- **P4** -- **Financial Fraud Detection:** Polytope Fraud Theory using supervised learning (*Int. Rev. Financial Analysis*, 2025)
- **P5** -- **AI in Economics:** Agentic workflows for economic research using LLMs and multimodal AI (arXiv)

Skills for Competition Economics

Methodological Expertise

- **Agent-Based Modeling:** Design, implementation (Python/Julia/R), and calibration of complex economic systems
- **Counterfactual Policy Analysis:** Evaluating policy interventions under heterogeneity and feedback mechanisms
- **Machine Learning:** Pattern detection in complex data using supervised learning (XGBoost, Neural Networks, Random Forest)
- **Empirical Calibration:** Simulated Method of Moments, neural network surrogates, and validation techniques

Research Methodology Framework

Empirical Rigor Across All Papers

Each research project follows a disciplined four-stage process:

1. **Hypothesis Formation:** Identify market failures, regulatory gaps, or strategic puzzles from theory and real-world observations
2. **Data Collection:** Gather empirical evidence from firm reports, regulatory databases, financial markets, or experimental simulations
3. **Statistical Analysis:** Apply econometric methods, machine learning, or agent-based modeling to test hypotheses
4. **Findings Interpretation:** Translate results into policy implications and strategic insights

Paper 2: Capacity Investment & Network Effects (JMP)

Hypothesis Formation

RQ: How do service capacity constraints, congestion, and consumer networks jointly shape firm competition in ICT markets?
Theory: Network effects amplify user-base advantages, but congestion degrades quality unless firms invest sufficiently in capacity

Data Collection

- ICT market characteristics: network effects, congestion patterns, capacity constraints
- Consumer behavior: small-world network structure, information diffusion, switching decisions
- Firm strategies: pricing, physical R&D investment, capacity management

Paper 2: Capacity Investment & Network Effects (cont.)

Statistical Analysis

- Agent-based computational model with duopoly competition
- Network dynamics: small-world topology with information diffusion
- Firm heuristics: myopic profit-based investment adjustments

Findings Interpretation

- Capacity is a quality driver and differentiation tool, not just production constraint
- Network effects amplify market concentration; congestion limits dominant firm profitability
- Technological edge enables dominance; smaller firms compete via pricing flexibility
- Frequent investment $\neq$ higher profits; strategic timing matters

Paper 2: Network Structure

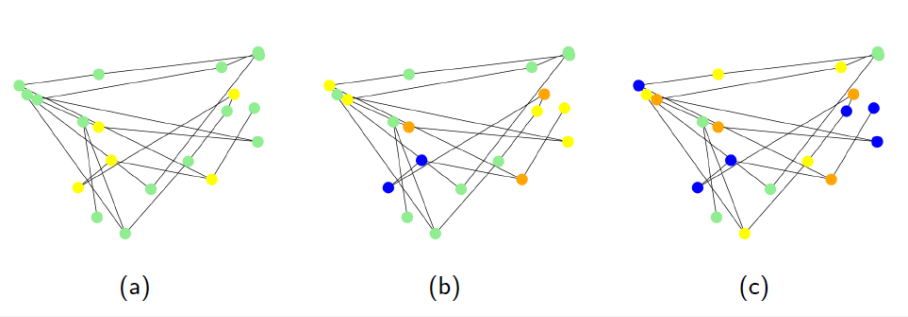


Figure 1: Small-world network structure enabling consumer information diffusion

Paper 2: Market Dynamics Simulation (1)

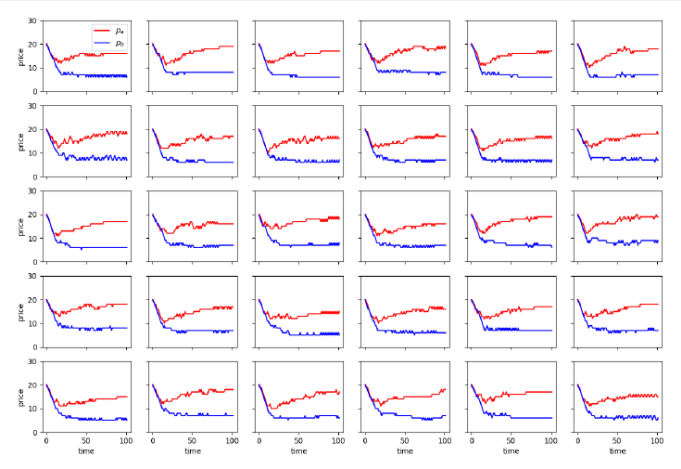


Figure 2: Time series of product prices

Paper 2: Market Dynamics Simulation (2)

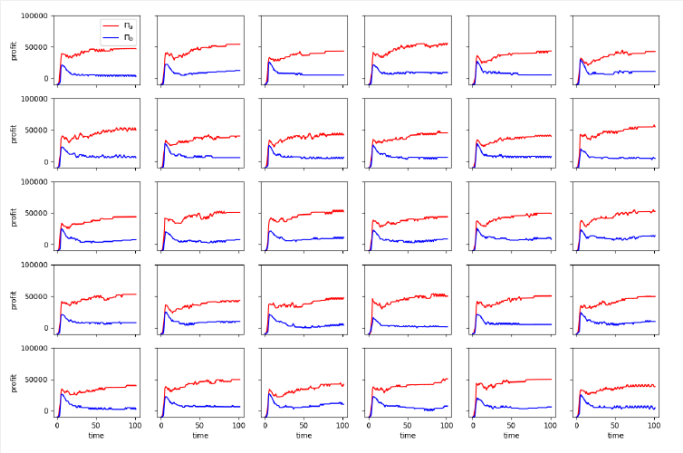


Figure 3: Time series of firm profits

Paper 1: Data Restrictions & Competition

Hypothesis Formation

RQ: How do varying levels of data privacy regulations influence price formation and market structures in software markets?
Theory: Data restrictions act as both compliance costs and product attributes (privacy-conscious consumers value restrictions)

Data Collection

- Stylized regional regulatory environments (low vs. high data restriction)
- Consumer preference distributions (functionality vs. privacy trade-offs)
- Firm cost structures (fixed R&D investment, variable compliance costs)

Paper 1: Data Restrictions & Competition (cont.)

Statistical Analysis

- Game-theoretic equilibrium analysis (two-stage investment-pricing game)
- Agent-based simulations with heterogeneous consumers (Hotelling framework)
- Counterfactual scenarios across regulatory regimes

Findings Interpretation

- Consumer taste distribution determines competitive outcomes: indifferent consumers $\rightarrow$ low-restriction firm dominates; privacy-conscious consumers $\rightarrow$ intense competition
- Cross-regional regulatory cooperation increases both consumer welfare and firm profits
- Regional differentiation in restrictions benefits both firms through product differentiation

Paper 3: ABM Calibration with Real Telecom Data

Hypothesis Formation

RQ: Which estimation techniques are most suitable for calibrating agent-based models to real-world data?
Theory: ABM credibility requires empirical grounding; parameter estimation remains a major methodological challenge

Data Collection

- U.S. telecom sector: AT&T vs. T-Mobile (2019-2023)
- Firm-level data: revenue, subscribers, costs, profits (monthly frequency)
- Sources: Official firm reports, Statista, Refinitiv
- Temporal disaggregation: annual/quarterly $\rightarrow$ monthly

Paper 3: ABM Calibration with Real Telecom Data (cont.)

Statistical Analysis

- Simulated Method of Moments (SMM): minimize NRMSE between simulated and empirical moments
- Neural Network Surrogate: maps 21 parameters $\rightarrow$ loss function; enables gradient-based optimization
- Latin Hypercube Sampling for parameter space exploration (16 target parameters)

Findings Interpretation

- SMM and NN surrogate are complementary: SMM matches moments; NN enables efficient re-use and gradient search
- Price-adjustment parameter identified as most influential for firm control
- Calibrated model reproduces observed market dynamics, confirming empirical relevance

Paper 4: Polytope Fraud Theory

Hypothesis Formation

RQ: Can financial statement fraud be reliably detected using public accounting data and machine learning?
Theory: Activist short sellers identify fraud patterns; these signals can be formalized as financial red flags

Data Collection

- Manually labeled dataset: activist short sellers' targets (fraudulent firms)
- Control group: non-fraudulent firms matched by industry and size
- Financial metrics: 10 red flags derived from short-seller behavior (cash flow anomalies, revenue quality, etc.)
- Public financial statements from SEC filings

Paper 4: Polytope Fraud Theory (cont.)

Statistical Analysis

- Supervised learning: Random Forest, Neural Networks, XGBoost
- Feature engineering: 10 financial red flags as predictors
- Cross-validation and out-of-sample testing
- Model interpretation: feature importance analysis

Findings Interpretation

- XGBoost achieves strong performance: 72% recall, 82% F1-score
- Ten financial red flags reliably identify fraudulent firms
- Published in *International Review of Financial Analysis* (2025)
- Investors and regulators can use these signals for early fraud detection

Paper 5: Agentic Workflows in Economic Research

Hypothesis Formation

RQ: Can Large Language Models and multimodal AI automate tasks across the economic research lifecycle?
Theory: Specialized AI agents with human-in-the-loop oversight can enhance efficiency and reproducibility

Data Collection & Implementation

- Framework design: agentic workflow combining LLMs and multimodal AI
- Implementation: Microsoft AutoGen with specialized agents
- Tasks: literature review, data analysis, empirical modeling, visualization
- Human oversight: validation checkpoints at each stage

Paper 5: Agentic Workflows in Economic Research (cont.)

Statistical Analysis

- Multi-agent system design with task decomposition
- Workflow orchestration: sequential and parallel agent execution
- Quality control: human validation at critical decision points

Findings Interpretation

- Agentic workflows can automate routine research tasks while maintaining rigor
- Human-in-the-loop oversight essential for methodological validity
- Enhances reproducibility and efficiency in economic research
- Demonstrates practical implementation

Summary: Why This Matters for Competition Economics

- **Methodological toolkit:** ABM + ML + empirical calibration
- **Policy-relevant insights:** Regulation, market structure, firm strategy
- **Proven track record:** Published research with real-world applications
- **Ready to contribute:** Applying the methods to real competition cases